

# Note on: *Pricing Total Return Swap*

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Reading notes

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## Metadata

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## One-paragraph summary

The paper resolves the post-2008 “TRS pricing dilemma” (Libor discounting double-counts upfront funding; firm-funding discounting breaks the law of one price) by separating two channels through which the cost of hedge financing enters: the *security leg* carries the repo financing cost of the underlying via a stock-rate drift  $r_s - q$ , and the *funding leg* continues to be discounted at the collateral / OIS rate. A self-financing wealth equation derived from the dealer’s hedge portfolio yields a single nonlinear PDE (3) whose discount rate  $r_e$  switches based on the sign of the unsecured exposure  $W = V - L$ . For fully collateralised TRS, the analytic solution (6) shows the funding premium discounted at  $r$  (OIS) and the asset leg reduced by  $\text{fva} = (\exp(\int_t^T (r_s - r) \, du) - 1) S_t$ , the repo-spread cost of hedging. Under repo-style margin (collateral follows the bond price minus accrued interest), (9) gives a tractable analytic form when  $r_b = r_c$ . For uncollateralised or partially collateralised trades, a recursive binomial / trinomial tree (Sec. 3) with switching discount rate solves the BSDE; total CRA decomposes into cva, dva, cfa, dfa (11). The framework extends to defaultable bond underliers (12) where the security-leg fva acquires terms for coupon and recovery savings. Numerical tables provide tight tree-vs.-analytic benchmarks (relative error  $\sim 10^{-13}$  at full collateral,  $\sim 10^{-4}$  at repo-style margin with 250 daily steps).

## 1 Setup and notation

| Paper              | House             | Meaning                                                        |
|--------------------|-------------------|----------------------------------------------------------------|
| $S_t$              | $S_t$             | Reference security (stock or bond) market price                |
| $M_0$              | —                 | Funding notional ( $= S_0$ typically)                          |
| $r_f$              | —                 | TRS funding rate $= l + s_f$ (e.g. Libor + spread)             |
| $s_f$              | —                 | Fixed funding premium spread                                   |
| $r$                | $r^c$             | Risk-free / OIS rate                                           |
| $r_s$              | $r^{\text{repo}}$ | Repo rate on the underlying security ( $r_s \geq r$ )          |
| $r_b, r_c$         | —                 | Senior unsecured bond rates of dealer (B) / counterparty (C)   |
| $r_w$              | —                 | $r_b \mathbf{1}_{W \leq 0} + r_c \mathbf{1}_{W > 0}$           |
| $r_N$              | —                 | Dealer's debt issuance rate                                    |
| $r_e$              | —                 | Effective derivative discount rate (5)                         |
| $q$                | —                 | Continuous dividend yield                                      |
| $L_t$              | —                 | Cash collateral posted                                         |
| $W_t$              | —                 | Unsecured exposure $W_t = V_t - L_t$                           |
| $\mu_t$            | —                 | Collateral-to-PV ratio $\mu_t = L_t/V_t$                       |
| $h$                | —                 | Repo haircut                                                   |
| $\Delta_t$         | —                 | Hedge ratio $-\partial V/\partial S$                           |
| $\lambda$          | —                 | Underlying default intensity (Sec. 5)                          |
| $R$                | Rec               | Recovery rate                                                  |
| $\Gamma_t$         | —                 | Joint survival indicator (1 at time $t$ if both survive)       |
| fva                | —                 | Funding value adj. to the asset leg (lower-case: TRS-specific) |
| cva, dva, cfa, dfa | —                 | XVA components from credit / funding basis split               |
| $\tilde{V}_t$      | $V_t^*$           | Counterparty-risk-free / OIS-discounted PV                     |
| $U_t$              | —                 | $V_t^* - V_t$ , total counterparty risk adjustment (CRA)       |

## 2 Key equations

**Pre-crisis baseline (Libor discounting):**

$$V(t) = \mathbb{E}_t^Q \left[ e^{-\int_t^T r \, du} (M_0 r_f (T - t_0) - (S_T - S_0)) \right] = S_0 (1 + r_f (T - t_0)) D_{t,T} - S_t, \quad (1)$$

(non-dividend,  $M_0 = S_0$ ). *Says:* the textbook valuation. At-the-issue,  $V_0 = s_f (T - t_0) S_0 D_{0,T}$  — the PV of the *whole* future funding premium, which under a CSA forces an upfront collateral posting equal to that PV. This is the repo trader's complaint.

**Self-financing wealth equation** (Sec. 2.2, eq. (6) in paper):

$$\begin{aligned} dM_t = & rM_t dt + (1 - \Gamma_t) [dW_t - r_w W_t dt + dN_t - r_N N_t dt + dL_t - rL_t dt \\ & + \Delta_t S_t q dt - d\Delta_t (S_t + dS_t) + dL_t^s - r_s L_t^s dt], \end{aligned} \quad (2)$$

with  $L_t^s = \Delta_t S_t$  the stock-repo account. *Says:* pre-default, the dealer holds: a hedge in  $\Delta_t$  shares financed at  $r_s$  in the repo market, posts/receives collateral  $L_t$  earning  $r$ , finances the unsecured exposure  $W_t$  at  $r_w$ , and rolls debt  $N_t$  at  $r_N$ . Net cash flows into the deposit account  $M_t$  earning  $r$ .

**The pricing PDE (collateralised, with hedge financing):**

$$\frac{\partial V}{\partial t} + (r_s - q) S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rL - r_w(V - L) = 0, \quad (3)$$

or equivalently with  $\mu = L/V$ ,

$$\frac{\partial V}{\partial t} + (r_s - q) S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - (\mu r + (1 - \mu) r_w) V = 0. \quad (4)$$

*Says:* **drift** = **repo** ( $r_s - q$ , the cost of carry of the hedge), **discount** = **effective rate**  $r_e$  on the RHS. The discount rate switches on the sign of  $W$ , hence on the sign of  $V - L$ .

**Effective derivative discount rate:**

$$r_e(t) = \mu r + (1 - \mu) [r_b(t) \mathbf{1}_{W(t) \leq 0} + r_c(t) \mathbf{1}_{W(t) > 0}]. \quad (5)$$

*Says:* Feynman-Kac gives  $V(t) = \mathbb{E}_t^Q[\exp(-\int_t^T r_e du) H(T)]$ . At full CSA ( $\mu = 1$ ),  $r_e = r$ ; at zero collateral ( $\mu = 0$ ),  $r_e = r_w$  which switches between dealer and counterparty funding rates.

**Fully collateralised TRS (closed form):**

$$V(t) = (M_0 r_f (T - t_0) + S_0) D_{t,T} - S_t e^{\int_t^T (r_s - r) du}. \quad (6)$$

*Says:* the funding premium  $M_0 r_f (T - t_0)$  and the initial security price  $S_0$  are both discounted at the OIS rate  $r$ ; the asset leg is the current price  $S_t$  multiplied by the repo-cost compounding factor. The OIS discounting of the funding premium contradicts the trader-funding-rate FVA argument (Fries-Lichtner) and resolves the dilemma in favour of the repo-channel interpretation.

**Repo-financing fva on the asset leg:**

$$\text{fva} = (e^{\int_t^T (r_s - r) du} - 1) S_t \geq 0, \quad (7)$$

since  $r_s \geq r$ . *Says:* this is a deterministic adjustment to the asset leg PV, distinct from the capital-F FVA reserved for uncollateralised swaps. Equivalently:  $\text{fva} = (F - S_t) -$  (textbook value), where  $F = S_t e^{\int_t^T (r_s - q) du}$  is the equity forward at the repo rate.

**Repo-style margin (Sec. 2.4):**

$$L_t = S_0 - S_t + r_F M_0 (t - t_0). \quad (8)$$

*Says:* collateral follows the price-return ( $S_0 - S_t$ ) plus the accrued interest of the period. Lets the funding-leg discounting “de facto disappear” on the variation-margin side, mirroring how repos accrue funding periodically.

**Closed form under repo-style margin,  $r_b = r_c$ :**

$$V(t) = -S \left( \frac{r_b - r - (r_s - r) D_{T, r_b - r_s}}{r_b - r_s} \right) + (S_0 - t_0 M_0 r_f) \left( 1 - r \frac{1 - D_{T, r_b}}{r_b} \right) + M_0 r_f \left[ \frac{T D_{T, r_b} r}{r_b} + \left( t + \frac{1 - D_{T, r_b}}{r_b} \right) \right] \quad (9)$$

where  $D_{T,x} = e^{-\int_t^T x du}$  is a placeholder discount. *Says:* explicit but unwieldy; useful as a tree-validation target.

### Total counterparty-risk adjustment (CRA):

$$U(t) \equiv V^*(t) - V(t) = \mathbb{E}_t^Q \left[ \int_t^T (r_w - r) (V^*(s) - L(s)) e^{-\int_t^s r_w du} ds \right], \quad (10)$$

with  $V^*$  the OIS-discounted (counterparty-risk-free) value. Decomposes into four parts:

$$U_t = cva_t - dva_t + cfa_t - dfa_t, \quad (11)$$

with each term in (10) restricted by an indicator on  $\{V > 0\}$  or  $\{V \leq 0\}$  and multiplied by a credit spread  $s_c$  or  $s_b$  (CDS-implied) or a funding-basis spread  $\mu_c$  or  $\mu_b$ , where  $r_c - r = s_c + \mu_c$ ,  $r_b - r = s_b + \mu_b$ .

### Defaultable underlying (Sec. 5, intensity $\lambda$ ):

$$V(t) = M_0 r_f T \bar{D}(t, T) + B_0 \left( \bar{D}(t, T) + \mathbb{E}_t^Q \left[ \int_t^T \lambda e^{-\int_t^\tau (r+\lambda) du} d\tau \right] \right) - B_t - fva, \quad (12)$$

with risky DF  $\bar{D}(t, T) = \mathbb{E}_t^Q [e^{-\int_t^T (r+\lambda) du}]$  and  $\bar{r}_s = (1-h)r_s + hr_N$  the all-in funding cost under haircut  $h$ . *Says*: the funding premium is now *risky-discounted* (survival-weighted); the asset leg gains coupon-savings and recovery-savings terms in the fva (13).

### Bond-fva:

$$fva = \gamma(t, T) B_t - \sum_{i: T \geq T_i > t} c_i \bar{D}(t, T_i) \gamma(t, T) - \mathbb{E}_t^Q \left[ \int_t^T \lambda R \gamma(\tau, T) e^{-\int_t^\tau (r+\lambda) du} d\tau \right], \quad (13)$$

with  $\gamma(t, T) = e^{\int_t^T (\bar{r}_s - r) du} - 1$ . *Says*: three contributions — (i) repo-spread cost on current bond price, (ii) savings on coupons received before  $T$ , (iii) savings on recovery payment on default.

## 3 Derivations & commentary

### Commentary

The headline resolution: **funding leg discounted at OIS, asset leg charged the repo spread**. This is the clean statement of where the cost of carry actually enters. Compared with Burgess's single-risky-DF framework, Lou's setup is more explicit: there is no single "risky DF" shared by both legs; the repo cost is a modification of the security-leg drift, not of the discount factor. At full CSA the two views can be reconciled, but partial / no CSA breaks any single-rate description.

### Commentary

The PDE (4) is structurally the same as the *Anonymous Discounting* note's eq. (8): drift uses  $r^{\text{repo}}$  (Lou's  $r_s$ ); discount uses an effective rate  $\mu r + (1 - \mu)r_w$  that under perfect CSA collapses to  $r$ . The  $\mu$ -formulation (4) makes the "how much of the exposure is collateralised" parameter explicit, which is the right level of abstraction for partial-CSA modelling.

### Commentary

**The nonlinearity.** The effective discount rate  $r_e$  in (5) depends on the sign of  $W = V - L$ . So  $V$  depends on  $r_e$  which depends on  $V$ . This makes pricing a BSDE, not a linear PDE. The tree-model trick (Sec. 3) is to fix the sign of  $W$  at the previous time step before rolling backward — decoupling discount rate and fair value within each step. This is the same device used in Lou (2017) for uncollateralised options.

### Commentary

The numerical tables are useful anchors: Table 1 shows the tree matches the analytic full-CSA formula to  $\sim 10^{-13}$  on six test cases. Table 2 shows repo-style margin convergence in number of steps. Table 3 contrasts the no-CSA payer and receiver values at asymmetric funding rates ( $r_b = 12\%$ ,  $r_c = 15\%$ ), showing a  $-0.4068$  bid-ask gap dominated by cva-dva asymmetry. Repo-style margin reduces XVAs by an order of magnitude.

### Commentary

The defaultable-bond extension (Sec. 5) is the part most directly useful for fixed-income work. The fva now contains three terms (13): the cost on the current price, savings from coupons that get repo-financed only until they're paid, and savings from the recovery payment that arrives before TRS maturity (in expectation). Operationally, the all-in repo cost  $\bar{r}_s = (1 - h)r_s + hr_N$  packages haircut into a single effective rate.

## 4 Validation

### Validation

This is a *specification* of mathematical objects the paper defines and the numerical values they should take. It says nothing about how those objects are implemented or invoked.

#### Quantities the paper defines:

- Pre-crisis TRS value (1).
- The pricing PDE (3)/(4) and its effective discount rate  $r_e$  (5).
- Closed-form fully-collateralised TRS PV (6).
- Hedge-financing adjustment fva (7).
- Closed-form repo-style margin PV (9) (under  $r_b = r_c$ ).
- Total CRA (10) and its four-component decomposition (11).
- Defaultable-bond TRS PV (12) and bond-fva (13).

#### Canonical numerical cases (Tables 1–3):

*Case A — Tree vs. analytic, full CSA* (Table 1):  $S = 100$ ,  $\sigma = 50\%$ ,  $r = 10\%$ ,  $q = 0$ , repo spread = 0 (last column = 5%),  $T = 1$ .

| Case          | ATI 1p                 | ATI 1p $K = 80$ | 1 stub $t_0 = 0.25$ | ATI 4p     | 4p $t_0 = 0.1$ | 4p $t_0 = 0.1$ |
|---------------|------------------------|-----------------|---------------------|------------|----------------|----------------|
| Analytic      | 0                      | -1.9032516      | 2.5315121           | -0.3597770 | 0.7160658      |                |
| Binomial Tree | $1.42 \times 10^{-13}$ | -1.9032516      | 2.5315121           | -0.3597770 | 0.7160658      |                |

*Case B — Repo-style margin convergence* (Table 2):  $r_b = r_c = 2\%$ , 1Y ATI, repo spread 5%, other inputs as Case A.

| Time steps | 100         | 250         | 500         | 1000        | Analytic    |
|------------|-------------|-------------|-------------|-------------|-------------|
| NPV        | -0.52348862 | -0.52336251 | -0.52332021 | -0.52329901 | -0.52327778 |

Case C — Uncollateralised TRS XVAs (Table 3):  $S = 100$ ,  $K = 100$ ,  $\sigma = 0.5$ ,  $q = 0$ ,  $T = 1$ ,  $r = 0.1$ ,  $r_s = 0.02$ ,  $r_f = 12.131\%$ ,  $r_b = 0.12$ ,  $r_c = 0.15$ ; spreads  $s_b = 1.7\%$ ,  $s_c = 4.2\%$ ,  $\mu_b = 0.3\%$ ,  $\mu_c = 0.8\%$ .

| Quantity | Payer no CSA | Receiver no CSA | Payer repo margin | Receiver repo margin |
|----------|--------------|-----------------|-------------------|----------------------|
| dva      | 0.1199       | 0.1120          | 0.0137            | 0.0069               |
| cva      | 0.2760       | 0.2948          | 0.0168            | 0.0334               |
| dfa      | 0.0212       | 0.0198          | 0.0024            | 0.0012               |
| cfa      | 0.0526       | 0.0561          | 0.0032            | 0.0064               |
| $V^*$    | -0.5701      | 0.5701          | -0.5701           | 0.5701               |
| $V$      | -0.7577      | 0.3509          | -0.5742           | 0.5384               |

**Equation  $\leftrightarrow$  quantity map:**

| Paper eq. | Symbol                                                    | Quantity                                            |
|-----------|-----------------------------------------------------------|-----------------------------------------------------|
| (1)       | $V(t)$ (pre-crisis)                                       | textbook Libor-discounted TRS PV                    |
| (3)–(4)   | PDE on $V$                                                | pricing PDE with separate drift / discount          |
| (5)       | $r_e$                                                     | effective discount rate (switching on sign of $W$ ) |
| (6)       | $V$ (full CSA)                                            | closed-form fully-collateralised TRS PV             |
| (7)       | fva                                                       | repo-financing cost adjustment to asset leg         |
| (8)–(9)   | $L_t, V(t)$                                               | repo-style margin and resulting closed form         |
| (10)–(11) | $U_t = \text{cva} - \text{dva} + \text{cfa} - \text{dfa}$ | total CRA and its 4-way split                       |
| (12)–(13) | $V, \text{fva}$                                           | defaultable-bond TRS PV and bond-fva                |

**Invariants and identities to verify:**

1. *Pre-crisis day-1 value identity*: at-the-issue with  $r_f = l + s_f$ , the textbook formula gives  $V_0 = s_f(T - t_0) S_0 D_{0,T}$  exactly — the PV of the entire funding premium.
2. *Full-CSA  $\rightarrow$  pre-crisis identity*: when  $r_s = r$  (no repo special), (6) reduces to (1) adjusted for dividends; specifically  $\text{fva} \rightarrow 0$  and the asset leg is simply  $S_t$ .
3. *Reproduction of Table 1*: the closed-form (6) on the six test cases matches the analytic column to all digits printed ( $\sim 10^{-7}$ ); the tree is within  $\sim 10^{-13}$ .
4. *Repo-style margin convergence* (Table 2): as the number of time steps doubles, the tree NPV approaches the (9) analytic value  $-0.52327778$  quasi-linearly in  $\Delta t$ .
5. *Bid-ask gap from XVA asymmetry*: at  $r_b = 12\%$ ,  $r_c = 15\%$  (Table 3),  $V^{\text{payer}} - V^{\text{receiver}}$  equals minus the sum of  $2 \text{cva} - 2 \text{dva} + 2 \text{cfa} - 2 \text{dfa}$ .
6. *Repo-style margin reduces XVA by an order of magnitude*: each row of Table 3 has the repo-margin XVA roughly  $10\times$  smaller than the no-CSA value.
7. *Full CSA  $\implies$  zero CRA*: in (10)  $U \equiv 0$  when  $L = V^* = V$ .
8. *Defaultable bond limit*: as  $\lambda \rightarrow 0$ , (12) reduces to the non-defaultable full-CSA formula (6) with  $S = B$ , and (13) reduces to the simple equity-fva (7).
9. *Implied par premium*: setting  $V = 0$  and solving for  $r_f - l$  in the full-CSA formula gives a closed-form breakeven funding spread; for the worked Case C, fixing the repo spread at 2%, the payer fair premium is  $r_f - l = 12.9541\%$  and receiver 12.5136% (bid-ask 44 bp).

## 5 Glossary of symbols (paper's notation)

|                    |                                                                                  |
|--------------------|----------------------------------------------------------------------------------|
| $S_t$              | Reference security spot price                                                    |
| $M_0$              | Funding leg notional (often = $S_0$ )                                            |
| $T, t_0, t_i$      | Maturity, previous payment date, future payment dates                            |
| $r_f = l + s_f$    | Funding rate (Libor + spread); $l$ is the floating fixing, $s_f$ a fixed premium |
| $r$                | Risk-free / OIS short rate                                                       |
| $r_s$              | Repo rate on the underlying security ( $r_s \geq r$ )                            |
| $q$                | Continuous dividend yield                                                        |
| $r_b, r_c$         | Dealer / counterparty senior unsecured rates                                     |
| $s_b, s_c$         | CDS spread components of $r_b - r$ , $r_c - r$                                   |
| $\mu_b, \mu_c$     | Funding-basis components ( $r_b - r = s_b + \mu_b$ )                             |
| $r_w$              | $r_b \mathbf{1}_{W \leq 0} + r_c \mathbf{1}_{W > 0}$                             |
| $r_N$              | Dealer's rolled-debt rate                                                        |
| $r_e$              | Effective derivative discount rate                                               |
| $L_t$              | Cash collateral at $t$                                                           |
| $W_t$              | $V_t - L_t$ , unsecured exposure                                                 |
| $\mu_t$            | $L_t/V_t$ , collateral ratio                                                     |
| $\Delta_t$         | $-\partial V/\partial S$ , hedge ratio                                           |
| $h$                | Repo haircut                                                                     |
| $\bar{r}_s$        | $(1 - h)r_s + hr_N$ , all-in bond hedge funding cost                             |
| $\Gamma_t$         | Joint default indicator (0 pre-default, 1 post)                                  |
| $\lambda$          | Underlying default intensity (Sec. 5)                                            |
| $R$                | Bond recovery rate                                                               |
| $\bar{D}(t, T)$    | Risky DF $\mathbb{E}_t^Q[e^{-\int_t^T (r+\lambda) du}]$                          |
| fva                | Funding value adj. to asset leg (TRS-specific; lower case)                       |
| cva, dva, cfa, dfa | Credit/funding adjustment components                                             |
| $V_t^*$            | OIS-discounted counterparty-risk-free PV                                         |
| $U_t$              | $V_t^* - V_t$ , total CRA                                                        |
| $D_{T,x}$          | Discount factor $e^{-\int_t^T x du}$ for placeholder rate $x$                    |
| $\gamma(t, T)$     | $e^{\int_t^T (\bar{r}_s - r) du} - 1$ , repo funding cost factor                 |

### TODO / Open questions

- Reproduce Table 1 (full-CSA tree vs. analytic) on the six test cases. Convergence target:  $\sim 10^{-13}$  relative error.
- Reproduce Table 2 (repo-style margin convergence) with 100, 250, 500, 1000 steps; check linear-in- $\Delta t$  convergence toward  $-0.52327778$ .
- Reproduce Table 3 (uncollateralised XVAs) using the trinomial tree; verify the bid-ask gap and the order-of-magnitude reduction under repo-style margin.
- Equity forward consistency check:  $F = S_t e^{\int_t^T (r_s - q) du}$  should agree with the asset-leg-only term of (6) when  $r_f = 0$  and  $M_0 r_f (T - t_0) \rightarrow F$ ; i.e. a bond forward is a TRS with zero funding leg (mentioned at the end of §5).
- Defaultable-bond extension: build the  $\lambda$ -trinomial tree from Sec. 5 and price a bond forward as a TRS with  $r_f = 0$ , then solve  $V(t) = 0$  for  $F$ ; compare against an independent forward bond calculation under the same assumptions.
- Cross-check against Burgess: at full CSA, both frameworks should give the same PV. Identify the parameters under which the Burgess single-risky-DF treatment is exactly equivalent to Lou's split treatment.